

prabir@Inspiron-5570:~/Desktop/chipletsim\$./simulator --benchmark E3S-AutoIndust

Loading E3S Benchmark: E3S-AutoIndust	
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Parsing E3S file: benchmarks/e3s/auto-indust-cords.tgff

Entering @COMMUN table

Comm type 0 = 4000 bytes

Comm type 1 = 8000 bytes

Comm type 2 = 15000 bytes

Comm type 3 = 1000 bytes

Found TASK_GRAPH 0

Task: src (type 45)

Task: can1 (type 0)

Task: fp (type 1)

Task: can2 (type 0)

Task: pulse (type 12)

Task: sink (type 45)

Arc: src -> can1 (type 0)

Arc: can1 -> fp (type 0)

Arc: fp -> can2 (type 0)

Arc: can2 -> pulse (type 0)

Arc: pulse -> sink (type 1)

Completed TASK_GRAPH 0

Found TASK_GRAPH 1

Task: src (type 45)

Task: iir (type 7)

Task: idct (type 8)

Task: sink (type 45)

Arc: src -> iir (type 0)

Arc: iir -> idct (type 0)

Arc: idct -> sink (type 0)

Completed TASK_GRAPH 1

Found TASK_GRAPH 2

Task: src (type 45)

Task: fft (type 5)

Task: matrix (type 10)

Task: ifft (type 9)

Task: fir (type 6)

Task: angle (type 0)

Task: road (type 13)

Task: table (type 14)

Task: sink (type 45)

Arc: src -> fir (type 0)

Arc: fir -> angle (type 0)

Arc: src -> fft (type 2)

Arc: fft -> matrix (type 2)

Arc: matrix -> ifft (type 2)

Arc: ifft -> angle (type 2)

Arc: angle -> road (type 0)

Arc: road -> table (type 0)

Arc: table -> sink (type 3)

Completed TASK_GRAPH 2

Found TASK_GRAPH 3

Task: src (type 45)

Task: ptr (type 11)
Task: cache (type 3)
Task: tooth (type 15)
Task: sink (type 45)
Arc: src -> ptr (type 3)
Arc: ptr -> cache (type 1)
Arc: cache -> tooth (type 1)
Arc: tooth -> sink (type 3)
Completed TASK_GRAPH 3

Parsing complete:

Total tasks: 24
Total arcs: 21
Comm types: 4

Creating tasks...

[0] src (type:45, IO_BOUND, HIGH, 32500 cycles)
[1] can1 (type:0, COMPUTE_LIGHT, MEDIUM, 1000 cycles)
[2] fp (type:1, COMPUTE_LIGHT, MEDIUM, 1100 cycles)
[3] can2 (type:0, COMPUTE_LIGHT, MEDIUM, 1000 cycles)
[4] pulse (type:12, COMPUTE_LIGHT, MEDIUM, 2200 cycles)
[5] sink (type:45, IO_BOUND, HIGH, 32500 cycles)
[6] src (type:45, IO_BOUND, HIGH, 32500 cycles)
[7] iir (type:7, COMPUTE_LIGHT, MEDIUM, 1700 cycles)
[8] idct (type:8, COMPUTE_LIGHT, MEDIUM, 1800 cycles)
[9] sink (type:45, IO_BOUND, HIGH, 32500 cycles)
[10] src (type:45, IO_BOUND, HIGH, 32500 cycles)
[11] fft (type:5, COMPUTE_LIGHT, MEDIUM, 1500 cycles)
[12] matrix (type:10, COMPUTE_LIGHT, MEDIUM, 2000 cycles)
[13] ifft (type:9, COMPUTE_LIGHT, MEDIUM, 1900 cycles)
[14] fir (type:6, COMPUTE_LIGHT, MEDIUM, 1600 cycles)
[15] angle (type:0, COMPUTE_LIGHT, MEDIUM, 1000 cycles)
[16] road (type:13, COMPUTE_LIGHT, MEDIUM, 2300 cycles)
[17] table (type:14, COMPUTE_LIGHT, MEDIUM, 2400 cycles)
[18] sink (type:45, IO_BOUND, HIGH, 32500 cycles)
[19] src (type:45, IO_BOUND, HIGH, 32500 cycles)
[20] ptr (type:11, COMPUTE_LIGHT, MEDIUM, 2100 cycles)
[21] cache (type:3, COMPUTE_LIGHT, MEDIUM, 1300 cycles)
[22] tooth (type:15, COMPUTE_HEAVY, MEDIUM, 5500 cycles)
[23] sink (type:45, IO_BOUND, HIGH, 32500 cycles)

Creating edges...

src -> can1 (4000 bytes)
can1 -> fp (4000 bytes)
fp -> can2 (4000 bytes)
can2 -> pulse (4000 bytes)
pulse -> sink (8000 bytes)
src -> iir (4000 bytes)
iir -> idct (4000 bytes)
idct -> sink (4000 bytes)
src -> fir (4000 bytes)
fir -> angle (4000 bytes)
src -> fft (15000 bytes)
fft -> matrix (15000 bytes)
matrix -> ifft (15000 bytes)
ifft -> angle (15000 bytes)

angle -> road (4000 bytes)
road -> table (4000 bytes)
table -> sink (1000 bytes)
src -> ptr (1000 bytes)
ptr -> cache (8000 bytes)
cache -> tooth (8000 bytes)
tooth -> sink (1000 bytes)

✓ E3S benchmark loaded successfully!
Total tasks: 24
Total edges: 21

HexaMesh Chiplet Task Mapping Simulator	
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Benchmark: E3S-AutoIndust
Automotive/Industrial control (engine, ABS, transmission) (24 tasks from E3S)

Configuration:
Rings: 1 (7 chiplets)
Cores per chiplet: 4
Total cores: 28
Clustering alpha: 0.7
Clustering beta: 0.3

Running simulation...

[GRAPH-AWARE CLUSTERING] Detecting application graphs...

Found 4 application graphs:

Graph 1: 4 tasks
Graph 0: 6 tasks
Graph 2: 9 tasks
Graph 3: 5 tasks

[CLUSTERING] Processing graph 1 (4 tasks)
Graph 1 clustered into 4 clusters

[CLUSTERING] Processing graph 0 (6 tasks)
Graph 0 clustered into 6 clusters

[CLUSTERING] Processing graph 2 (9 tasks)
Graph 2 clustered into 9 clusters

[CLUSTERING] Processing graph 3 (5 tasks)
Graph 3 clustered into 5 clusters

[CLUSTERING] Total clusters: 8

Writing output files to: output/E3S-AutoIndust

✓ Output files created:

- summary.json (topology, results, per-chiplet stats)
- task_placement.csv (task details, placement, timing)
- comm_latency.csv (per-edge communication latency)

- core_utilization.csv (per-core statistics)
- event_log.csv (simulation events timeline)
- cluster_report.csv (cluster formation details)

SIMULATION RESULTS

Execution:

Makespan: 80566 cycles
Tasks placed: 24 / 24
Tasks deferred: 0
Clusters formed: 8

Resource Utilization:

Average chiplet utilization: 13.3493%
Average core utilization: 24.8767%
Load imbalance (σ): 0.0664341

Communication:

Total communication edges: 21
Average communication latency: 422.095 cycles
Min communication latency: 90 cycles
Max communication latency: 965 cycles

Fault Tolerance:

Faults detected: 0
Replicas created: 8

Performance:

Total wait cycles: 8587
Average task wait: 357.792 cycles

✓ Simulation complete!

Results saved to: ./output/E3S-AutoIndust/

```
python3 analyze_placement.py
```

```
=== Task Distribution by Chiplet ===
```

```
Chiplet 0: 5 tasks
```

```
Chiplet 1: 4 tasks
```

```
Chiplet 2: 5 tasks
```

```
Chiplet 3: 2 tasks
```

```
Chiplet 4: 4 tasks
```

```
Chiplet 5: 4 tasks
```

```
=== Task Distribution by Graph ===
```

```
Graph 0:
```

```
Chiplets used: [2, 4]
```

```
g0_src_t45 → Chiplet 4
```

```
g0_can1_t0 → Chiplet 4
```

```
g0_fp_t1 → Chiplet 4
```

```
g0_can2_t0 → Chiplet 4
```

```
g0_pulse_t12 → Chiplet 2
```

```
g0_sink_t45 → Chiplet 2
```

Graph 1:

Chiplelets used: [1]

g1_src_t45 → Chippet 1

g1_iir_t7 → Chippet 1

g1_idct_t8 → Chippet 1

g1_sink_t45 → Chippet 1

Graph 2:

Chiplelets used: [0, 5]

g2_src_t45 → Chippet 5

g2_fft_t5 → Chippet 5

g2_matrix_t10 → Chippet 5

g2_ifft_t9 → Chippet 5

g2_fir_t6 → Chippet 0

g2_angle_t0 → Chippet 0

g2_road_t13 → Chippet 0

g2_table_t14 → Chippet 0

g2_sink_t45 → Chippet 0

Graph 3:

Chiplelets used: [2, 3]

g3_src_t45 → Chippet 3

g3_ptr_t11 → Chippet 2

g3_cache_t3 → Chippet 2

g3_tooth_t15 → Chippet 2

g3_sink_t45 → Chippet 3

python3 analyze_comm.py

INTER-CHIPLET: g0_can2_t0 (C4) → g0_pulse_t12 (C2)

INTER-CHIPLET: g2_src_t45 (C5) → g2_fir_t6 (C0)

INTER-CHIPLET: g2_ifft_t9 (C5) → g2_angle_t0 (C0)

INTER-CHIPLET: g3_src_t45 (C3) → g3_ptr_t11 (C2)

INTER-CHIPLET: g3_tooth_t15 (C2) → g3_sink_t45 (C3)

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Intra-chippet: 16/21 (76.2%)

Inter-chippet: 5/21 (23.8%)

TARGET: Inter-chippet should be 20-40%